

## Technical Datasheet 4G Gateway

### Product Overview:-

The 4G Gateway is an advanced networking device designed to provide seamless connectivity and data management capabilities for various applications. With support for 4G LTE connectivity and TCP Ethernet, cloud connectivity via MQTT protocol, and robust storage options, it offers a comprehensive solution for remote monitoring and control systems. Equipped with Modbus RS485 port and micro Sim compatibility, it caters to a wide range of industrial and IoT applications. The intuitive web-based UI interface allows for easy configuration and management, making it an ideal choice for both professionals and enthusiasts.



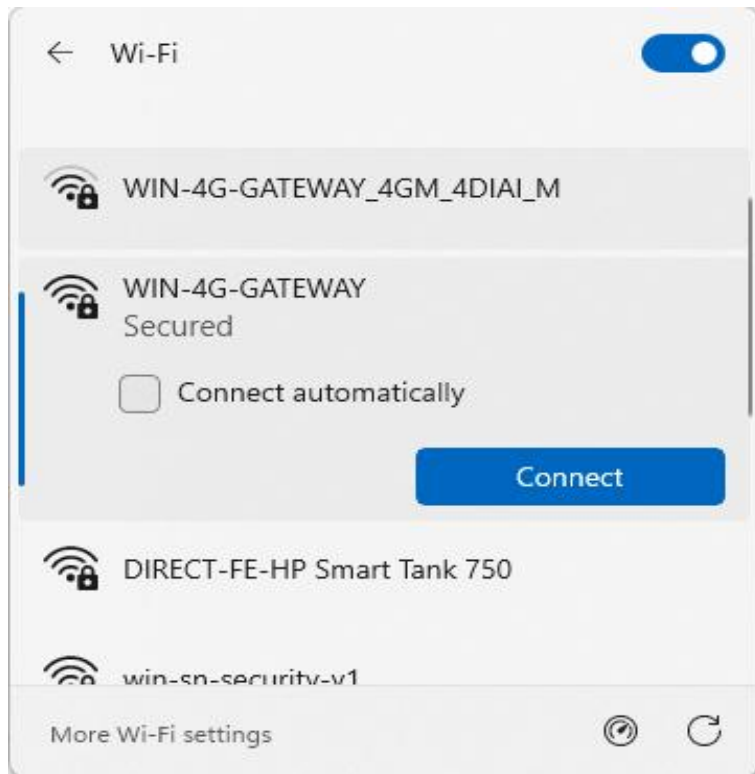
## Specifications

### General:-

- **4G LTE Connectivity:** Utilizes 4G LTE technology for reliable and high-speed internet access. If both 4G, Wi-Fi and Ethernet are unavailable, the data is securely stored in the SD card for later upload.
- **Cloud Connectivity (MQTT):** Seamlessly connects to cloud platforms using the MQTT protocol for efficient data transmission and management.
- **Event Storage:** Offers on board event storage, capable of storing up to 50,000 events or more. Includes an external memory card slot for easy data retrieval in the event of 4G network issues.
- **Modbus RS485 Port:** Features 1 port of Modbus RS485 for integration with industrial control systems and sensors.
- **Micro Sim Compatibility:** Supports micro-Sim card for flexible and convenient connectivity option.
- **Web-based UI Interface:** Intuitive web-based user interface for easy configuration, monitoring, and management.
- **Dimensions:** 110 mm L x 110 mm B x 50 mm H.
- **Power:** Input Power – 15-40 VDC, Typical – 12V DC @ 150mA.
- **Operating Temperature:** 0 – 60C (32 ~ 140F).
- **Storage Temperature:** 20 - 70C (-4 ~ 158F).
- **Storage Humidity:** 5 ~ 95 % RH, non – Condensing.

## LED Indicators:-

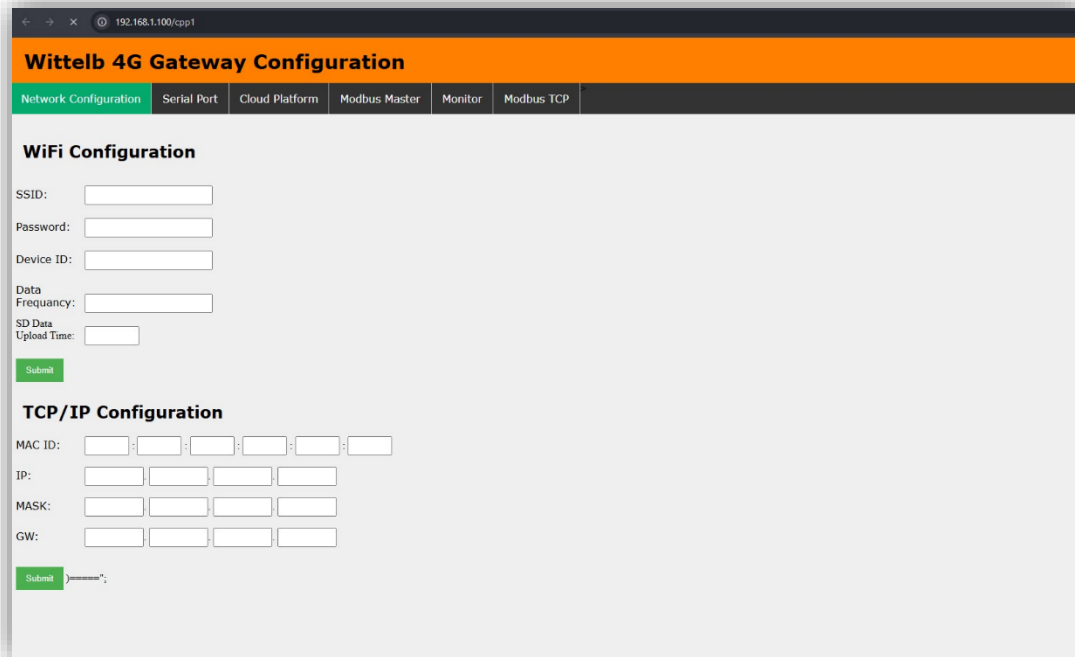
- **POWER:** Device power status.
- **NETWORK:** Indicates 4G/Wi-Fi network connectivity.
- **CLOUD:** Shows MQTT cloud connection status.
- **POLL:** POLL/WI-FI: Indicates Modbus polling activity / Wi-Fi mode
- **RX:** Data received from RS485.
- **TX:** Data transmitted to RS485.



## Configuration Settings:

- Power up the module by connecting a 12 V DC supply.
- Insert the SIM card into the designated slot.
- Place the Micro SD card into its designated slot.
- Connect MODBUS RTU Multiple Slave devices (up to 31) to the A & B Terminal Of Module.
- Access the module's configuration hotspot (ID: WIN-4G-GATEWAY, PASS: 123456789).
- Default IP Address of Module is "192.168.1.100/cpp1".
- Upon entering the provided IP address, a configuration webpage Pages will be accessible.

## Page 1: -



**Wittelb 4G Gateway Configuration**

Network Configuration | Serial Port | Cloud Platform | Modbus Master | Monitor | Modbus TCP

**WiFi Configuration**

SSID:

Password:

Device ID:

Data Frequency:

SD Data Upload Time:

**TCP/IP Configuration**

MAC ID:  :  :  :  :  :

IP:  .  .  .

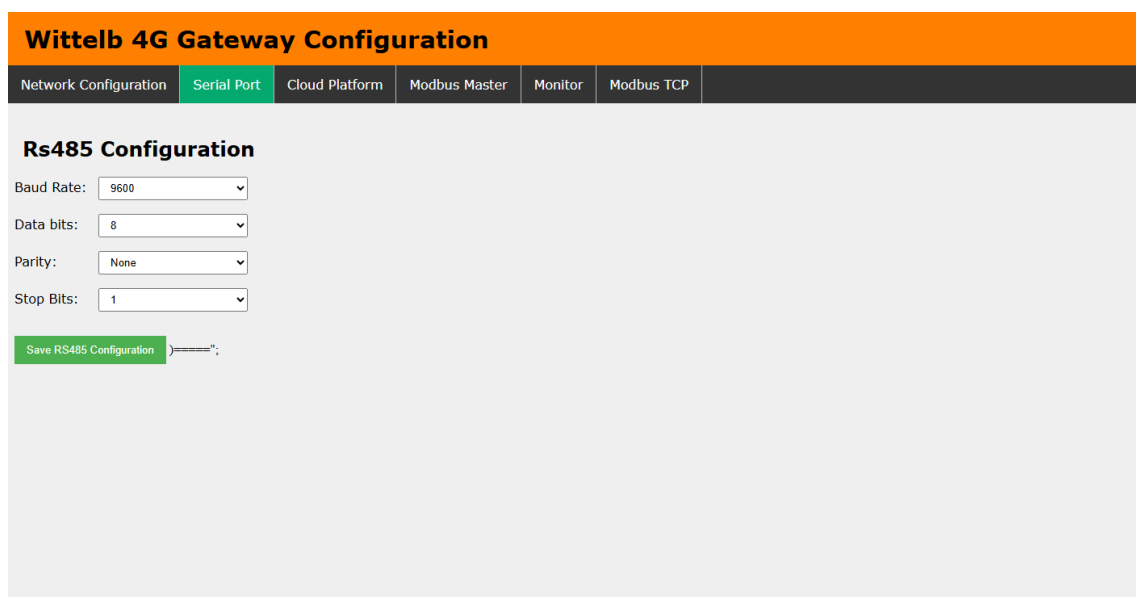
MASK:  .  .  .

GW:  .  .  .

- Enter Device ID (Should be unique for every device).
- Enter data frequency in min to receive data on cloud.
- Enter time to upload SD card data to cloud.

## Page 2:-

On second page you can set device Baud Rate (9600 / 19200 / 38400 / 57600 / 115200), Data Bits (5 / 6 / 7 / 8), Parity (None / Odd / Even), and Stop Bits (1 / 2) for RS485 Communication



**Wittelb 4G Gateway Configuration**

Network Configuration | **Serial Port** | Cloud Platform | Modbus Master | Monitor | Modbus TCP

**Rs485 Configuration**

Baud Rate:

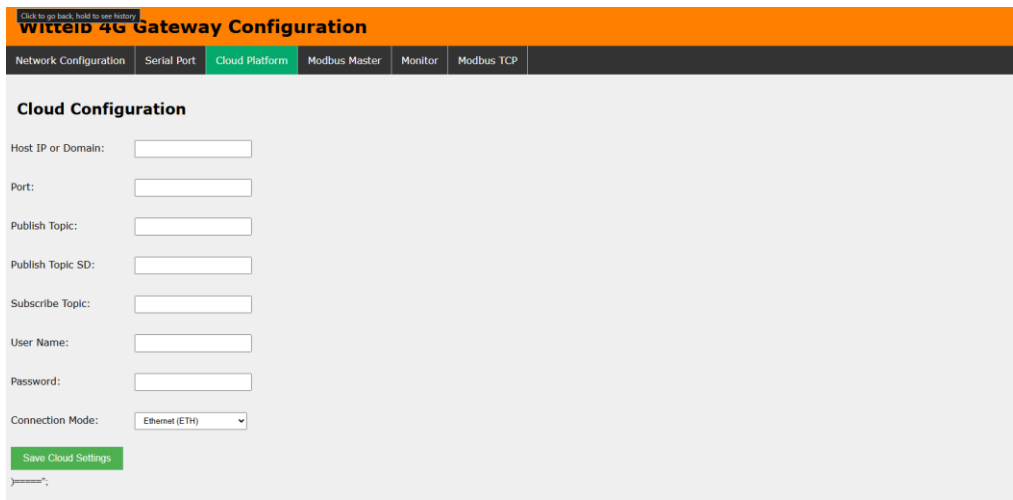
Data bits:

Parity:

Stop Bits:

### Page 3:-

- Navigate to page 3 on the cloud platform.
- Input the Host IP or Domain.
- Enter the port Number.
- Enter the topics for publishing and receiving messages.
- Provide a username and password if required.
- Allow a few seconds for the module to restart and establish a connection to MQTT.
- Select Wi-fi, TCP and 4G from Which Source you need to send data over MQTT



The screenshot shows the 'WitteIB 4G Gateway Configuration' web interface. The 'Cloud Platform' tab is selected, displaying a 'Cloud Configuration' section. This section includes input fields for 'Host IP or Domain', 'Port', 'Publish Topic', 'Publish Topic SD', 'Subscribe Topic', 'User Name', and 'Password'. A 'Connection Mode' dropdown menu is set to 'Ethernet (ETH)'. A green 'Save Cloud Settings' button is located at the bottom left of the configuration area.

### Page 4:-

- Go to page 4 on the Modbus Master interface.
- Input the name and slave address for the designated slave.
- Specify the register address from which you want to read slave data.
- You can enter only the Register which you need to read.
- Select the Data Type: UINT16, Float32 Big Endian, Float32 Little Endian, Float32 BE Byte Swap, or Float32 LE Byte Swap.
- Enter the function code corresponding to the desired slave.

**Wittelb 4G Gateway Configuration**

Network Configuration
 Serial Port
 Cloud Platform
 **Modbus Master**
 Monitor
 Modbus TCP
 About

### Modbus Master Configuration

| Name  | Slave Address | Register | Function Code | Data Type                                                                                                              | Actions |
|-------|---------------|----------|---------------|------------------------------------------------------------------------------------------------------------------------|---------|
| Slave | 1             | 12       | 3             | UNIT16<br><b>UNIT16</b><br>Float32 Big Endian<br>Float32 Little Endian<br>Float32 BE Byte Swap<br>Float32 LE Byte Swap | Delete  |

Add Row
 Submit

## Page5:-

- On page 5 (Monitor), view live Modbus register values for all configured entries.
- Table auto-refreshes every 7 seconds showing: Slave Address, Register, Function Code, Tag Name, Data Type, Live Value, and Status (OK / FAIL)
- Use Slave Address and Register filter fields to narrow down the live view.

**Wittelb 4G Gateway Configuration**

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### Modbus Live Monitor

Total Registers  
 --

Slaves Active  
 --

Showing  
 --

Slave Address  
 e.g. 1

Register  
 e.g. 100

Filter

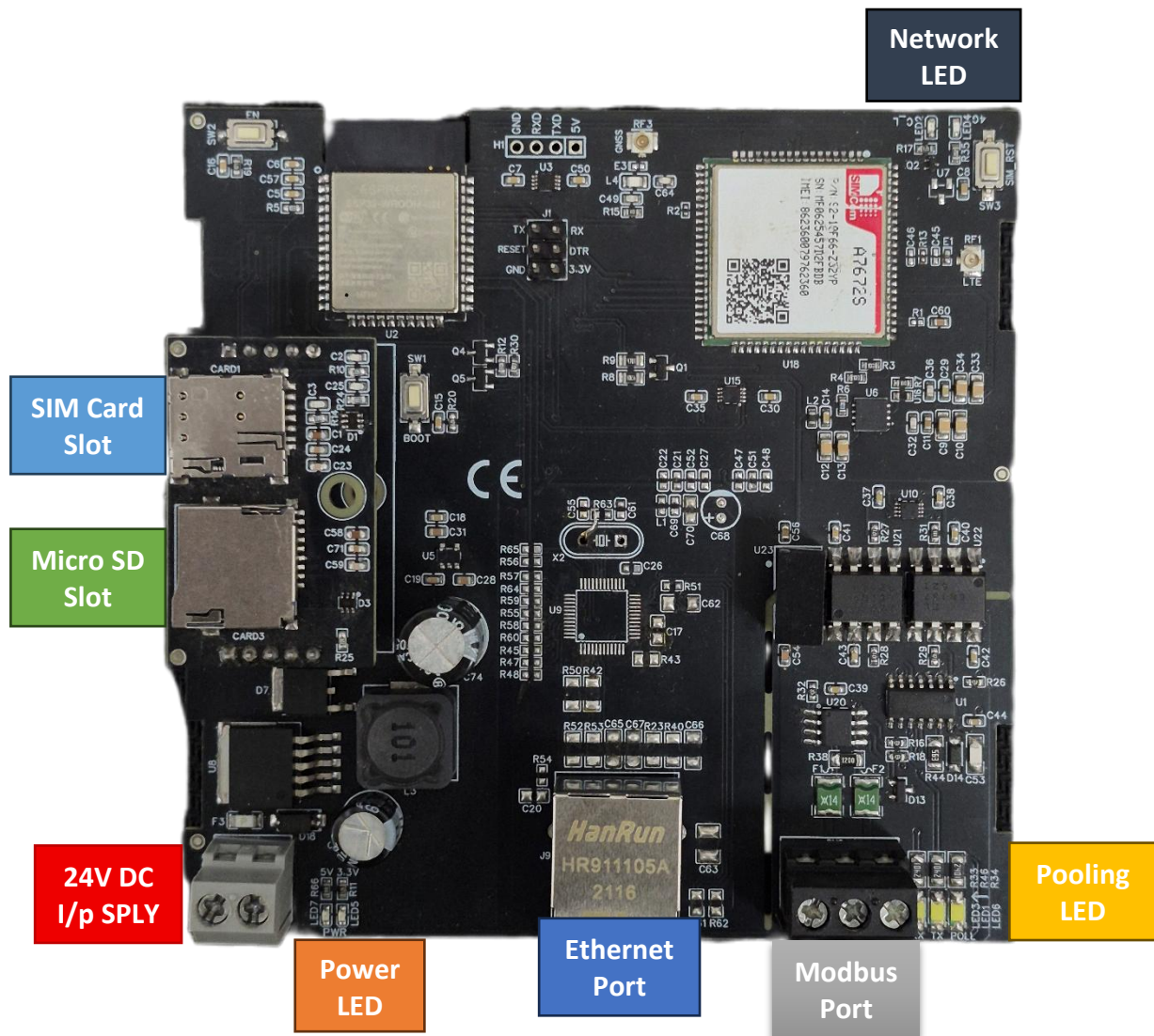
Clear

Connection failed — retrying...

Next refresh in 3s

| Sr.No           | Slave Address | Register | Function Code | Name / Tag | Data Type | Live Value | Status | Last Updated |
|-----------------|---------------|----------|---------------|------------|-----------|------------|--------|--------------|
| Loading data... |               |          |               |            |           |            |        |              |

## 4G Gateway connection Diagram: -



## Additional Features: -

Communication ports are isolated  
Input power reverse polarity safety  
ESD Safety IEC 61000-4-2,  $\pm 30\text{KV}$  contact,  $\pm 30\text{KV}$  air  
EFT IEC 61000-4-4, 50A (5/50ms) 400V isolation.

## Contact us: -

Augmatic Technologies Pvt. Ltd.,  
Plot no 6, Shah Industrial Estate II,  
Kotambi, Vadodara – 391510.

Email – [Sales@wittelb.com](mailto:Sales@wittelb.com)

Support: [support@wittelb.raiseaticket.com](mailto:support@wittelb.raiseaticket.com) / +91 94038 92805